

Electronic Companion

Neural Differential Utility: Endogenous Service Contracts and Certified Dynamic Optimization

Revision R4, September 20, 2026

This companion retains supporting derivations, counterexamples, implementation details, and the mapping of historical evidence. Equation and theorem references without an EC prefix refer to the main paper. The canonical operating model is the externally priced, perishable-inventory contract model of Section 2. Historical internal-penalty examples are retained as distinct models, not silently identified with that model.

EC.1 Comparator geometry and inner optimization

EC.1.1 What an exposed-face argument proves

Fix a state and a physical action, and let the continuation-value vector range over an open subset $U \subset \mathbb{R}^d$. A finite family of admissible valuation controls induces

$$F(v) = \max_{j=1,\dots,J} \{g_j + \beta p_j^\top v\}. \quad (\text{EC.1})$$

Here p_j is a transition probability vector and g_j includes the control-dependent payoff and adjustment terms. The restrictions on the comparator must be specified before any non-equivalence statement is made.

Proposition EC.1.1 (Two exposed slopes). *Suppose $\beta > 0$ and there are open subsets $U_1, U_2 \subset U$ on which distinct indices j_1, j_2 are the unique maximizers, with $p_{j_1} \neq p_{j_2}$. No single affine primitive $g + \beta p^\top v$ equals F throughout U . Nevertheless, an ordinary Markov decision problem with the valuation index admitted as an action represents F exactly.*

Proof. On U_i , the derivative of F is βp_{j_i} . Equality with a single affine function on an open set would require its derivative to equal both distinct vectors. This is impossible. The last statement follows by taking the ordinary action set to be the original physical action paired with j . $\square$

Different intercepts alone do not establish the premise: if all slopes coincide, the largest intercept gives a single affine envelope. Nor does the proposition say that a continuum of affine functions must have a nonsmooth envelope. For example, with strictly positive reference probabilities p_i and $\rho > 0$, set

$$Z = \sum_i p_i e^{\rho \beta v_i}, \quad q_i^* = p_i e^{\rho \beta v_i} / Z.$$

For any probability vector q ,

$$\beta q^\top v - \rho^{-1} \text{KL}(q \| p) = \rho^{-1} \log Z - \rho^{-1} \text{KL}(q \| q^*). \quad (\text{EC.2})$$

Nonnegativity of relative entropy proves the Gibbs variational identity. Entropic continuation is therefore an exact member of a sufficiently rich payoff–kernel envelope family. A fixed entropic risk parameter does not

exclude this representation. The sign of the risk parameter matters: the displayed maximization convention is for $\rho > 0$; the corresponding negative-parameter convention reverses the variational optimization. No general exclusion of risk-sensitive or robust control is claimed.

Exogenous preference processes and fixed multiplier formulations can be compared only after fixing their admissible transition/payoff families. A restricted family may fail to contain a particular joint policy; that is not a theorem that every conventional stochastic-control formulation fails. Expanded-action equality in Proposition 2.1 of the main paper is the definitive representation check.

EC.1.2 Conditional strongly concave reductions

A separate sufficient computational condition, useful in continuous-control models, is strong concavity of a particular inner Hamiltonian. It is not needed for the exact envelope algorithm or the quadratic contract-grid bound.

Proposition EC.1.2 (Inner-selector residual bound). *Let f be differentiable and μ -strongly concave on a compact convex set $\Theta \subset \mathbb{R}^d$. Let $\widehat{\theta} \in \Theta$ satisfy*

$$\nabla f(\widehat{\theta}) = n + e, \quad n \in N_{\Theta}(\widehat{\theta}), \quad \|e\| \leq \varepsilon,$$

where $N_{\Theta}(x) = \{n : n^{\top}(y - x) \leq 0 \ \forall y \in \Theta\}$. Then the unique maximizer θ^* satisfies

$$\|\theta^* - \widehat{\theta}\| \leq \varepsilon/\mu, \quad f(\theta^*) - f(\widehat{\theta}) \leq \varepsilon^2/(2\mu).$$

For $f(\theta) = b^{\top}\theta - \theta^{\top}R\theta/2$ with $R \succ 0$, the optimizer is the projection of $R^{-1}b$ onto Θ in the R norm. In the scalar interval case it is ordinary clipping.

Proof. Strong concavity and the normal-cone inequality give

$$f(\theta^*) - f(\widehat{\theta}) \leq \varepsilon d - \mu d^2/2 \leq \varepsilon^2/(2\mu), \quad d = \|\theta^* - \widehat{\theta}\|.$$

For the distance bound, combine the strong monotonicity of $-\nabla f$ with the first-order optimality condition at θ^* and the stated approximate normal-cone condition. They imply $\mu d^2 \leq \varepsilon d$. Completing the square in the quadratic objective proves the projection statement. $\square$

Thus a physical-action approximation with gap δ_a , combined with an inner residual ε , has total maximization gap at most $\delta_a + \varepsilon^2/(2\mu)$ when these hypotheses hold uniformly. Positive maintenance cost by itself does not establish global concavity after taking the stock envelope and adding continuation. The main paper avoids that unsupported inference.

EC.2 Physical metrics and the historical newsvendor argument

EC.2.1 Restricted critical-fractile mismatch

Consider the separate physical cost

$$C_z(S) = h\mathbb{E}(S - D_z)^+ + p_z\mathbb{E}(D_z - S)^+, \quad h, p_z > 0.$$

Assume the relevant stock interval has a density $f_z \geq \underline{f}_z > 0$ and the interior minimizer S_z^* satisfies $F_z(S_z^*) = p_z/(h + p_z)$. A fixed service parameter uses a common effective shortage coefficient $\bar{p}$ and selects $\bar{S}_z$ at fractile $\bar{p}/(h + \bar{p})$. Then

$$C_z(\bar{S}_z) - C_z(S_z^*) \geq \frac{(h + p_z)\underline{f}_z}{2}(\bar{S}_z - S_z^*)^2. \quad (\text{EC.3})$$

Indeed $C'_z(S_z^*) = 0$ and $C''_z = (h + p_z)f_z$ on the connecting interval; integrate the second derivative twice. A positive-weight collection of regimes yields a strict expected gap whenever at least one selected stock differs from its physical optimum. Bounds on density and on quantile separation make this a stable mismatch statement.

This argument is useful and is retained. Its comparator is a common critical-fractile parameter, not an unrestricted state-aware inventory controller. Selecting a state-dependent parameter afterward so that its penalty equals p_z does not prove that an endogenous net-reward optimizer selects it. The actual-optimal-policy accounting theorem in Section 4 supplies that missing distinction. In the new experiment the joint policy's physical cost is computed from its own exact optimizing actions, not from a physical-cost-matching contract inserted by hand.

EC.2.2 The actual optimized ledger

Write $W^\pi = K^\pi - C^\pi$, with all entries discounted under the same law. The exact results give

$$W^J - W^F = 0.944747060878850\dots, \quad C^F - C^J = 0.213203951173630\dots$$

Their difference is the increase in net contract transfers. The physical-cost oracle attains approximately 8.506094529; the joint policy costs 9.374173780. Thus improved net reward does not imply physical optimality. The oracle's contract-one implementation is fixed by the recorded tie convention, and its net cash flow is not the objective it minimizes.

Fill rate is discounted filled demand divided by discounted demand, not a reward rescaling. It declines from 0.9259259259... under the frozen contract to 0.8821088614... under the joint contract. This decline is part of the result, not a failed run to discard. The full output also retains undiscounted filled, lost, and total demand. Historical columns named `avg_cost` that include adjustment or valuation-dependent terms remain total-cost diagnostics; they are not relabeled as physical cost.

EC.3 Continuous-time derivations and boundary controls

EC.3.1 Value volatility and the adjoint

For shared Brownian noise, the general relation is

$$Z = \sigma_c^\top P_c + \sigma_h^\top P_u,$$

not two independently invertible observations. When $\sigma_h = 0$, the internal gradient is absent from this volatility. If the external block also lacks full row rank, some components of P_c are unobserved as well. If the physical action affects internal drift, the term $P_u^\top \partial_a h$ enters the physical first-order condition; the degeneracy need not be confined to the contract action.

For completeness, suppose the value is sufficiently smooth, diffusion is control-independent, the optimal selector satisfies the Hamiltonian envelope condition, and all required derivatives and stochastic integrals exist. Before exit, set $P = \nabla V$ and $Q = D^2 V \Sigma$. Differentiating the HJB equation and applying Itô's formula gives

$$dP_s = - \left[b_x^\top P_s + r_x + \sum_j (\sigma_j)_x^\top Q_{s,j} - \rho P_s \right] ds + Q_s dB_s. \quad (\text{EC.4})$$

Here derivatives of the primitives hold the control fixed; the envelope condition removes derivatives of the maximizing selector. Along paths reaching the terminal face without prior exit, $P_T = \nabla G$. A Dirichlet exit payoff fixes the boundary value and tangential derivatives when smooth, but does not fix the normal derivative. Consequently one must not impose $P_\tau = \nabla \psi$ in all directions merely from the Dirichlet value condition. The stopped value-process equation, rather than such an extra adjoint terminal condition, is used in the main certificate.

Adding an independent diffusion $\sqrt{2\varepsilon}I$ to the internal block gives $Z_u^\varepsilon = \sqrt{2\varepsilon}P_u$. Its inverse has squared amplification $1/(2\varepsilon)$ under additive volatility error. A direct gradient avoids this inverse but must still solve the value or adjoint problem. The main paper's least-squares parity result tests naturally generated data rather than adding a fixed error floor to the volatility target.

EC.3.2 Why both boundary faces and both comparisons are necessary

A terminal condition alone does not determine a compact spatial problem. On $x \in [1, 2]$, let

$$w(t, x) = (1 - t)^{-1/2} \exp\{-x^2/[4(1 - t)]\}, \quad t < 1, \quad w(1, x) = 0.$$

It satisfies $w_t + w_{xx} = 0$ and extends smoothly to zero at the terminal face on this spatial interval, yet it is nonzero in the interior. The zero function has the same terminal data and interior equation. Their distinct lateral values explain the difference. An interior and terminal residual cannot control that difference without boundary information.

A separate one-step example shows why a policy bound needs both critic comparisons. There are two actions leading deterministically to distinct terminal states, with true terminal reward zero. Action rewards

are 0 and 1.9. A terminal critic equal to +1 at the first state and -1 at the second has terminal sup error one. Its greedy action selects the first action, yielding true policy loss 1.9. The preceding critic can be its exact Bellman update, so the Bellman residual and greedy gap relative to that critic are zero. A policy bound of only one terminal error is false; two terminal errors suffice. This example is executed as an exact rational regression.

The main theorem separately bounds value error and the critic-to-actor value error. For an actor-induced residual, the difference between the actor and optimal Hamiltonians must also be charged. A Neumann defect, reflected boundary local time, and a state-constraint viscosity inequality are not interchangeable with the Dirichlet mismatch used here.

EC.3.3 From finite validation to a uniform bound

Suppose an independently established Lipschitz constant L_R bounds the interior residual over the specified compact domain, and validation nodes form a cover of fill distance h_R . Then

$$\sup_D |R| \leq \max_{x_i} |R(x_i)| + L_R h_R.$$

Terminal and lateral defects require their own covers and constants. An actor gap likewise requires a certified action cover and Hamiltonian continuity modulus. Training loss, a sampled percentile, or a ratio against another network supplies none of these constants by itself. A differentiable network does not automatically supply a useful global residual bound: derivative growth and domain coverage must be bounded.

A vanishing-viscosity value conclusion can be made under explicit comparison, common boundary attainment, uniform boundedness and equicontinuity, and locally uniform convergence of the HJB operators. Compactness gives subsequential value limits; stability of the sub- and supersolution inequalities identifies a solution of the limiting boundary problem; comparison identifies all such limits. Uniform value convergence alone does not prove convergence of gradients or recovery of the missing shadow price. These conditional observations preserve the original regularization motivation without using it as a substitute for the absolute certificate.

EC.4 Execution details and constructive neural evidence

EC.4.1 Frozen inputs and exact arithmetic

The experiment reads the original R2 protocol and verifies its SHA-256 before computing. The protocol bytes are unchanged. R3's recorded conventions specify time indices, the oracle's contract-one tie, duplicate gate labels, paired batch inference, and identical-data least squares. The contract-grid theorem, smooth neural compilation, and joint space-time-action study are additional analytic or deterministic calculations; they do not replace any recorded method or change its parameters.

For contract grid denominator n , write $S = s/2$, $\theta = j/n$, and inherited $q = i/n$. Let $w_d = (1, 3, 4, 2)$

and $D = z + d$. Define integer quantities

$$O = \sum_d w_d(s - 2D)^+, \quad L = \sum_d w_d(2D - s)^+, \quad C_{\text{num}} = 60s + 12O + 13L.$$

Expected physical cost is $C_{\text{num}}/400$. Expected net stage reward, on denominator $400n^2$, has numerator

$$80(3 + 7z)jn - C_{\text{num}}n^2 - 24jLn - 600j^2 - 240(j - i)^2. \quad (\text{EC.5})$$

The transition numerator matrix on denominator 20 is

$$\begin{pmatrix} 14 & 5 & 1 \\ 3 & 14 & 3 \\ 1 & 5 & 14 \end{pmatrix}, \quad \beta = 97/100.$$

With r periods remaining, values use common denominator $400n^22000^{r-1}$. Thus all action comparisons use integers with no numerical tie tolerance. A separately written implementation reconstructs expected costs from the demand realizations using `Fraction` arithmetic and enumerates the actions independently. It checks all 264 state–time values and policies for each of the five methods. It also verifies that the tie-selected pair is componentwise greatest among all maximizers.

The primary expanded-action comparator uses a flattened list of 143 physical-contract pairs. Its equality with the joint solver is exact at every state, not merely agreement after rounding. The dummy gate enumerates 11 duplicate labels per stock choice, but the economic contract and next inherited state stay at one half. It therefore has 13 distinct economic choices, not 143. Its output equals the frozen-contract solution exactly.

The 416 neighboring-state monotonicity checks support the displayed instance; the general result is the proof in the main paper. The minimum observed adjacent cross difference of the exact value is $9/250$. The exact upper hull is evaluated at all 264 primary states. The hull weights and rational switch locations are stored in `exact_envelopes.json`.

EC.4.2 Simulation and uncertainty

Each of 64 independent batches contains 4,096 episodes. A NumPy PCG64 generator is seeded with $2026092300 + b$, $b = 0, \dots, 63$. Demand and regime uniforms are generated before all policy evaluations, and the same exogenous histories are used by all methods within a batch. Confidence intervals use the two-sided Student quantile with 63 degrees of freedom applied to the batch means; comparisons use paired batch differences. They quantify simulation error, not training variability or external validity. No simulated sign determines which methods are retained.

The paired net-reward increase over the frozen contract is approximately 0.943729, with 95% interval $[0.939821, 0.947636]$. The exact expectation is $0.9447470609 \dots$. The paired physical-cost difference, joint minus frozen, is approximately -0.212112 , with interval $[-0.214933, -0.209291]$. The exact difference is $-0.2132039512 \dots$. All batch ledgers, not only aggregate intervals, are retained.

EC.4.3 Identical-data least-squares parity

The conditioning batches use seed 2026092400 + b and 2,048 pairs (u, ξ) . The same pairs are reused across the four noise levels within a batch. Let $X = [u^2/2, 1]$ and $X_Z = [u^2/(2\sqrt{2\varepsilon}), 1]$. For nonzero ε ,

$$X_Z = X \operatorname{diag}((2\varepsilon)^{-1/2}, 1).$$

Their fitted function spaces coincide. With full column rank, unregularized least squares gives $\hat{\zeta} = \sqrt{2\varepsilon} \hat{k}$ and the same intercept in exact arithmetic. Hence the inferred shadow-price functions coincide. Conditional expectation of the observed payoff is $u^2/2 + \varepsilon$, so the true gradient is u . Integrated error on uniform test states is $(\hat{k} - 1)^2/3$. The naturally induced volatility error is 2ε times this error. These identities explain why the experiment does not realize a fixed volatility-target error floor.

EC.4.4 Smooth neural weights and certified policies

For each state–time envelope, take its first affine line and its successive nonnegative slope jumps at the rational switch locations. Replace each positive-part hinge by the three cubic-ReLU units in Corollary 3.5 of the main paper. The quadratic skip term is $-\lambda q^2$. Thus every weight, shift, and coefficient is specified by `exact_envelopes.json` and the chosen rational τ . The network is C^2 in the inherited contract; time and regime index separate networks in this finite-horizon model. This is not a $C^{1,2}$ space–time solution of the diffusion problem.

The maximum network size is 27 cubic-ReLU units. For all five smoothing levels, exact rational policy evaluation verifies all 264 primary grid states, for 1,320 state-level checks. The policy does not change at any of these states. This exact finite-grid result is stronger than a small sampled training residual, but it does not assert policy equality at every off-grid initial contract. The uniform analytic policy bound covers those initial states and decreases as follows:

m in $\tau = 2^{-m}$	Uniform critic bound	Uniform policy-loss bound
4	1/80	0.077606934
8	1/1280	0.004850433
12	1/20480	0.000303152
16	1/327680	0.000018947
20	1/5242880	0.000001184

The critic bound follows from the exact triangular smoothing inequality, not from sampling. The policy bound is relative to the unchanged primary contract grid. Adding the contract-discretization budget gives a bound relative to continuous contracts on the same stock set. No post hoc target noise, optimizer selection, or claim of large-neural training is involved.

EC.4.5 Diffusion and mesh results

The manufactured stopped diffusion, its full parabolic payoff, and its perturbation sequence are specified in the main paper. The residual bound $2.30e + 1.05e^2$ is analytic. Terminal and lateral defects are identically zero.

The actor is analytically greedy and feasible throughout the domain. The additional explicit upwind/central scheme uses a nonnegative transition stencil under the displayed CFL condition, exact boundary data, and the actual refined state and time grids. Its numerical node errors and analytic node bounds are different columns. The latter does not certify an interpolated continuous-state actor. The earlier fixed-state inventory action-cover refinement is not used as evidence of space–time convergence.

EC.5 Preservation and interpretation of historical material

EC.5.1 An immutable historical snapshot

The entire source tree at reviewed commit

a91b707883b2408de58d9392e4fc546e08a00881

is retained as `archive/pre-r4/`. This is a Git-tree identity, not a reconstructed selection of files. The reviewed root manuscript has blob

2569eee2eb7f80f0a5b31ed07be75d5da9e6a0d3

and remains there unchanged, together with its PDF, bibliography, figures, code, tables, reviews, and release package. The new root manuscript and electronic companion are the active R4 text. The archive is provenance, not a second set of current theorem claims.

The main theoretical threads are retained with the following explicit treatment. Frozen-valuation embedding is now exact policy-class restriction and expanded-action equivalence. Exposed affine faces remain a comparator-specific result in EC.1, while entropic overlap is derived rather than excluded. Inventory structure is proved for the external-contract primitive; the penalty-only reversal is preserved as an exact counterexample. Strongly concave inner selection remains a conditional computational lemma, alongside the stronger envelope calculation that needs no such concavity. Physical critical-fractile mismatch remains in EC.2, while the actual optimizer is analyzed by the cash-flow identity. FBSDE/value-gradient relationships, adjoint scope, and vanishing-noise conditioning remain in EC.3. The HJB argument is replaced by a boundary-complete stopped-domain proof and a scheme-specific bound. The constructive neural representation and exact finite-model checks are new positive results.

EC.5.2 Historical numerical evidence is not discarded or upgraded

The earlier experiment families, SKU and exact-DP replays, uncertainty tables, actor–critic ablations, residual audits, figures, and manifests remain in the snapshot. They concern their original simulators and objectives. They are neither rerun by the R4 script nor pooled with the new contract experiment. Their successful software checks do not establish hypotheses of a theorem for a different model.

For example, the archived large-neural inventory ladder reports a native-HJB ratio of 0.192341487123, normalized-RMSE ratio 0.979300855398, terminal-loss ratio 0.929461677903, greedy-gap 95th percentile

29.326960415, and held-out increment 0.000199407352943. These numbers are retained as sampled diagnostics. The ladder’s literal label `sampled_inventory_gap_closure_pass` is not treated as a mathematical predicate. It provides neither an interior sup enclosure nor a lateral-boundary defect. The absence of those quantities is not replaced by a zero in any R4 certificate.

The archived finite inventory cover has 122 states and 1,220 Bellman nodes. Its nested action cover refines actions while retaining those state–time nodes. That is a finite-model action refinement, not evidence that a continuous state–time mesh tends to zero. Earlier direct-*P*/learned-*Z* ablations retain their architecture, optimizer, data, and target-error interpretation; the new identical-data least-squares calculation is a separate result and reports parity.

EC.5.3 Software governance outside the theorem stack

Hash consistency, package replay, release parity, validation guards, and reviewer handoff indexes are valuable reproducibility engineering. They are preserved in the historical tree and indexed in the R4 response and validation record. They are not restated as propositions proving scientific novelty or correctness. The new source entry point, executed result files, table generator, build commands, and artifact hashes make the present claims inspectable without an internal “referee closure” theorem. A failed or unverified scientific hypothesis cannot be converted to a certificate by changing a manifest label.

The revision therefore preserves the original material while giving each result a specific current role. It replaces incorrect statements with proofs and executable counterchecks rather than suppressing the history or asserting that all historical neural diagnostics have become certified.